

Paul Jerabek

Group Leader “Metal Hydride Modeling”

Deputy Head of Department of Materials Design

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Education

- Dec 2010 – Sep 2014 **Dr. rer. nat.**, *Philipps-Universität Marburg*
Thesis: *Clusterartige und hochkoordinierte Metallkomplexe*
- Jun 2010 – Nov 2010 **Dipl.-Chem.**, *Philipps-Universität Marburg*
Thesis: *Bindungstheoretische Untersuchungen der Systeme des Typs YE-X-EY*
- Oct 2005 – Jun 2010 **Studies of Chemistry**, *Philipps-Universität Marburg*

Professional experience

- Sep 2022 – present **Deputy Head of Department of Materials Design**, *Helmholtz-Zentrum Hereon*
Department administration and representation, third-party funding acquisition, research strategy planning, digital transformation.
- Oct 2021 – present **Co-Leader, German-New Zealand Green Hydrogen Centre (He Honoka Hauwai)**, *Helmholtz-Zentrum Hereon / University of Otago*
Administration and representation, strategic planning, networking, funding acquisition, event organization.
- Sep 2019 – present **Group Leader “Metal Hydride Modeling”**, *Helmholtz-Zentrum Hereon*
First-principles molecular dynamics, CALPHAD and phase-field simulations for the characterization and optimization of novel metal hydride materials for large-scale hydrogen storage. Close collaborations with Lawrence Livermore National Laboratory (USA) and the University of Otago (New Zealand).
- Jul 2018 – Aug 2019 **Humboldt Return Fellow**, *Max-Planck-Institut für Kohlenforschung*
Theoretical investigation of physical properties of organonitrogen; substituent and solvent effects on switching temperature in Fe(II) spin-crossover complexes; reaction mechanism exploration for asymmetrical, homogeneous catalysis; bonding situation and electronic properties of pincer complexes.
- Dec 2017 – Jun 2018 **Postdoctoral Research Assistant**, *Massey University*
Monte-Carlo melting simulations of heavy and superheavy elements; first-principles molecular dynamics simulations of coinage metals in liquid and solid phase; highly accurate calculations of atomic polarizabilities.
- Dec 2015 – Nov 2017 **Feodor Lynen Research Fellow**, *Massey University*
Photochemistry of small molecules in different environments; influence of relativistic and solvent effects on atomic and molecular bonding properties; development of highly accurate ab-initio many-body potentials for heavy and superheavy elements.
- Dec 2014 – Nov 2015 **Postdoctoral Research Assistant**, *Philipps-Universität Marburg*
Method development (C) for core electron spectroscopy of organic molecules on surfaces; density functional theory based investigation of transition metal clusters and complexes.

Dec 2010 – Nov 2014	Research Assistant, <i>Philipps-Universität Marburg</i> Quantum chemical investigation of main-group and transition metal compounds in collaboration with experimental groups in Bochum, München, Göttingen, Dresden, Münster and San Diego. Redesign of the lab course in Theoretical Chemistry.
Jan 2007 – Dec 2010	Student Assistant, <i>Philipps-Universität Marburg</i> Supervision of lab courses in Physical Chemistry and Chemistry for Medical Students.

Third-party funding

2024	Sustainable TiFe Generation from Secondary Sources BMBF (Germany) Principal Investigator 1.5 MEUR 3 years
2024	HIDA Visiting Researcher Grant (with S. Faouri) Helmholtz Association Host 10 kEUR 2 months
2023	HIDA Visiting Researcher Grant (with A. Rowberg) Helmholtz Association Host 15 kEUR 3 months
2022	Generating Hydrogen Storage Materials from New Zealand Resources BMBF (Germany) Coordinator 400 kEUR 3 years
2021	Establishing a German-New Zealand Research Centre for Green Hydrogen BMBF (Germany) Coordinator 760 kEUR 5 years
2021	HIDA Visiting Scientist Grant (with M. Selzer) Helmholtz Association Fellow 10 kEUR 3 months
2017	Humboldt Return Fellowship Alexander von Humboldt Foundation Fellow 47 kEUR 1 year
2016	Computation of highly accurate interatomic potentials of heavy and superheavy elements New Zealand eScience Infrastructure (NeSI) Principal Investigator 117 kEUR 1.5 years
2014	Hochpräzise quantenchemische Simulationen von atomaren und molekularen Phasen unter extremen Bedingungen Alexander von Humboldt Foundation Fellow 85 kEUR 2 years

Conference organization

2025	E-MRS Fall Meeting 2025 — Symposium A Co-organizer, Warsaw, Poland
2024	E-MRS Fall Meeting 2024 — Symposium C: Sustainable materials for chemical and electrochemical energy storage II Main organizer, Warsaw, Poland
2023	Inaugural New Zealand Hydrogen Symposium Co-organizer, Dunedin, New Zealand
2015	Bonding Analysis Workshop Conceptualization and main organizer, Marburg, Germany

Teaching

2025 SoSe	Hydrogen Technology Co-lecturer, Technische Universität Hamburg (TUHH), Hamburg
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2023 – present	Lectureship Lecturer, Technische Hochschule Mittelhessen, Gießen
2019 – 2019	Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa (Volkswagen-Stiftung) Tutor, University of Mauritius, Reduit
2014 – 2019	Hands-on workshops on electronic structure methods Workshop lecturer, Various universities, Hyderabad, Dunedin, Denton, Mariapfarr, Mauritius
2010 – 2014	Practical course in Theoretical Chemistry Teaching assistant, Philipps-Universität Marburg, Marburg
2007 – 2010	Physical Chemistry and Chemistry for Medical Students Student assistant, Philipps-Universität Marburg, Marburg

Talks and posters

May 2025	German-New Zealand WTZ Meeting 2025 , Wellington, New Zealand Invited Talk
Feb 2025	11th International Conference on Advanced Materials and Nanotechnology (AMN11) , Christchurch, New Zealand Invited Talk
Jul 2024	New Zealand-Germany Energy Dialogue 2024 , Berlin, Wellington, Germany, New Zealand Invited Talk
Feb 2023	10th International Conference on Advanced Materials and Nanotechnology (AMN10) , Rotorua, New Zealand Invited Talk
Jul 2018	18th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2018) , Rota, Spain Invited Talk
Jul 2016	16th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2016) , Rota, Spain Invited Talk
Mar 2024	German Physical Society (DPG) Spring Meeting 2024 , Berlin, Germany Contributed Talk
Jan 2024	2nd New Zealand Hydrogen Symposium (NZHS-2) , Wellington, New Zealand Contributed Talk
Mar 2023	German Physical Society (DPG) Spring Meeting 2023 , Dresden, Germany Contributed Talk
Sep 2022	European Materials Research Society (E-MRS) Fall Meeting 2022 , Warsaw, Poland Contributed Talk
Mar 2019	Chemiedozententagung 2019 , Koblenz, Germany Contributed Talk
Nov 2017	17th Biennial Symposium of the Australian and New Zealand Associations of von Humboldt Fellows , Wellington, New Zealand Contributed Talk
Sep 2017	12th International Conference on Relativistic Effects in Heavy-Element Chemistry and Physics (REHE) , Marburg, Germany Contributed Talk

Aug 2016	New Zealand Institute of Chemistry Conference (NZIC-16) , Queenstown, New Zealand Contributed Talk
May 2024	18th International Symposium of Metal-Hydrogen Systems (MH 2024) , Saint-Malo, France Poster
Sep 2018	54th Symposium on Theoretical Chemistry (STC) , Halle (Saale), Germany Poster
Aug 2017	11th World Association of Theoretical and Computational Chemists (WATOC) , Munich, Germany Poster
Jul 2016	9th Congress of the International Society for Theoretical Chemical Physics (ISTCP IX) , Grand Forks (ND), USA Poster
Sep 2015	51st Symposium on Theoretical Chemistry (STC) , Postdam, Germany Poster
Sep 2013	49th Symposium on Theoretical Chemistry (STC) , Erlangen, Germany Poster
Sep 2012	48th Symposium on Theoretical Chemistry (STC) , Karlsruhe, Germany Poster
Oct 2011	Modeling of Molecular Properties Conference (MolMod) , Heidelberg, Germany Poster
Jul 2011	9th World Association of Theoretical and Computational Chemists (WATOC) , Santiago de Compostela, Spain Poster
Sep 2010	46th Symposium on Theoretical Chemistry (STC) , Münster, Germany Poster
Feb 2025	University of Waikato , Hamilton, New Zealand Invited Seminar
Jun 2024	University of Hamburg , Hamburg, Germany Invited Seminar
Feb 2024	Massey University , Auckland, New Zealand Invited Seminar
Feb 2024	University of Waikato , Hamilton, New Zealand Invited Seminar
Mar 2023	Massey University , Auckland, New Zealand Invited Seminar
Mar 2023	University of Auckland , Auckland, New Zealand Invited Seminar
Dec 2021	Karlsruhe Institute of Technology (KIT) , Karlsruhe, Germany Invited Seminar
May 2019	Helmholtz-Zentrum für Material- und Küstenforschung , Geesthacht, Germany Invited Seminar
Apr 2019	Universität Graz , Graz, Austria Invited Seminar
Feb 2019	Technische Universität Berlin , Berlin, Germany Invited Seminar

Jan 2019	GSI Helmholtzzentrum für Schwerionenforschung , Darmstadt, Germany Invited Seminar
Feb 2018	University of Otago , Dunedin, New Zealand Invited Seminar
Jun 2017	University of Otago , Dunedin, New Zealand Invited Seminar
Jul 2016	University of North Texas , Denton (TX), USA Invited Seminar
Oct 2019	Summer School/Workshop on Computational Chemistry “Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa” , Reduit, Mauritius Workshop Lecture
Mar 2019	XIIth Workshop on Modern Methods in Quantum Chemistry , Mariapfarr, Austria Workshop Lecture
Dec 2017	University of Otago , Dunedin, New Zealand Workshop Lecture
Jul 2016	University of North Texas , Denton (TX), USA Workshop Lecture
Jan 2014	MCBR Winter School , Hyderabad, India Workshop Lecture

Peer-reviewed journal articles

70. “Evaluation of the synergetic effects of Ni and Cr in $\text{TiFe}_{(1-x-y)}\text{Cr}_x\text{Ni}_y$ for enhanced hydriding thermodynamics”: E. Pericoli, V. Ferretti, E. Alvares, P. Jerabek, C. Pistidda, L. Pasquini, *Adv. Energy Sustain. Res.*, **7**, e70243, 2026.
69. “Relativistic effects are crucial for uraniumphosphorus bonding - a DFT study of cyclopentadienyluranium-phosphine complexes”: J.-E. Ahrens, P. Jerabek, L. Vondung, *Mol. Phys.*, e2611400, 2026.
68. “Role of Elemental Substitution on Hydrogen Incorporation in Characteristic Ti-Fe-Based Oxides”: A. J. E. Rowberg, S. Kang, K. Sellschopp, P. Jerabek, T. W. Heo, B. C. Wood, *ACS Appl. Energy Mater.*, **9**, 1242–1253, 2026.
67. “Leveraging atomic disorder to modulate hydrogen storage thermodynamics in compositionally complex intermetallics”: Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Interdiscip. Mater.*, **0**, 1-13, 2026.
66. “Predicting hydrogen storage capacity loss of Si-containing FeTi alloys through ab initio thermodynamic modeling”: L. Dinachandran, E. Alvares, K. Sellschopp, T. Klassen, P. Jerabek, A. L. Garden, *Acta Mater.*, **301**, 121514, 2025.
65. “TiAlNb alloy interatomic potentials - comparing passive and active machine learning techniques with MTP and DeepMD”: A. Chandran, A. Santhosh, P. Jerabek, R. C. Aydin, C. J. Cyron, *Front. Mater.*, **12**, 2025.
64. “Low-temperature regeneration of metal hydride after impurity gas exposure”: Y. Shang, A. Santhosh, M. Ante, L. Wimbert, P. Jerabek, T. Schupp, T. Klassen, C. Pistidda, *Acta Mater.*, **298**, 121398, 2025.
63. “High Ionic Conduction in Rb- and Cs-Mixed Cation Amide for Energy Storage”: T. T. Le, K. Sellschopp, F. Murgia, A. L. Garden, S. Bordignon, J. P. Embs, M. R. Chierotti, A. Schökel, F. Karimi, P. Jerabek, T. Klassen, C. Pistidda, *Small*, 2502943, 2025.

62. "A Thermodynamic Approach to Model Multicomponent FeTi-based Alloys for Hydrogen Storage": E. Alvares, A. J. E. Rowberg, K. Sellschopp, B. C. Wood, T. Klassen, P. Jerabek, C. Pistidda, *Scr. Mater.*, **259**, 116516, 2025.
61. "Multiscale Modeling the FeTi Metal-Hydride Interphase Boundaries – Quantification of Decoupled Chemo-Mechanical Energies": E. Alvares, K. Sellschopp, B. Wang, S. Kang, T. Klassen, T. W. Heo, T. Klassen, P. Jerabek, C. Pistidda, *npj Comput. Mater.*, **10**, 249, 2024.
60. "Comparative analysis of ternary TiAlNb interatomic potentials - moment tensor vs. deep learning approaches": A. Chandran, A. Santhosh, C. Pistidda, P. Jerabek, R. Aydin, C. J. Cyron, *Front. Mater.*, **11**, 1466793, 2024.
59. "Thermodynamic assessment of the Ce-H and CeNi₅-H system": P. Hannappel, E. Alvares, F. Heubner, C. Pistidda, P. Jerabek, T. Weißgärber, *Calphad*, **85**, 102701, 2024.
58. "Predictable Electronic Tuning of FeII versus RuII Complexes via Choice of Azine Donor - An Electrochemical and Theoretical Study": M. Robb, L. Bondi, S. Rodriguez-Jimenez, A. Garden, P. Jerabek, S. Brooker, *Dalton Trans.*, **53**, 1999-2007, 2024.
57. "Tipping the balance between the bcc and fcc phase within the alkali and coinage metal groups": A. Robles-Navarro, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **63**, e202313679, 2024.
56. "First-principles study on interfacial property in MgB₂-based reactive hydride composites": Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Scr. Mater.*, **240**, 115837, 2024.
55. "Low-Cost, Multifunctional, and Sustainable Sodium Sulfido Ferrate(II)": M. Ghazanfari, K. Siemensmeyer, A. Santhosh, J. Vrijmoed, M. Tallu, S. Dehnen, P. Jerabek, G. Thiele, *Inorg. Chem.*, **62**, 15358-15366, 2023.
54. "Ultra-lightweight compositionally complex alloys with large ambient-temperature hydrogen storage capacity": Y. Shang, Z. Lei, E. Alvares, S. Garroni, R. Dore, M. Rustici, S. Enzo, A. Schökel, Y. Shi, P. Jerabek, Z. Lu, T. Klassen, C. Pistidda, *Mater. Today*, **67**, 113-126, 2023.
53. "Influence of near-surface oxide layers on TiFe hydrogenation - Mechanistic insights and implications for hydrogen storage applications": A. Santhosh, S. Kang, N. Keilbart, B. C. Wood, T. Klassen, P. Jerabek, M. Dornheim, *J. Mater. Chem. A*, **11**, 18776-18789, 2023.
52. "Experimental and Computational Studies on the Formation of Mixed Amide-Hydride Solid Solutions for CsNH₂-CsH System": T.-T. Le, A. Santhosh, S. Bordignon, M. R. Chierotti, P. Jerabek, T. Klassen, C. Pistidda, *Results in Engineering*, **17**, 100895, 2023.
51. "Magnesium- and intermetallic alloys-based hydrides for energy storage: modelling, synthesis and properties": L. Pasquini, K. Sakaki, E. Akiba, P. Jerabek et al., *Prog. Energy*, **4**, 32007, 2022.
50. "Relativistic Effects Stabilize Unusual Gold(II) Sulfate Structure via Auophilic Interactions": P. Jerabek, A. Santhosh, P. Schwerdtfeger, *Inorg. Chem.*, **61**, 13077-13084, 2022.
49. "Solving a problem with a single parameter: A smooth bcc to fcc phase transition for metallic lithium": P. Jerabek, A. Burrows, P. Schwerdtfeger, *Chem. Commun.*, **58**, 13369-13372, 2022.
48. "Remarkable Infrared falsenlinear Optical, Dielectric, and Strong Diamagnetic Characteristics of Semiconducting K₃[BiS₃]": M. R. Ghazanfari, L. Vittadello, D. Al-Sabbagh, A. Santhosh, C. Frankcom, F. Fuß, C. A. von Randow, K. Siemensmeyer, J. C. Vrijmoed, F. Emmerling, P. Jerabek, M. Imlau, G. Thiele, *J. Phys. Chem. Lett.*, **13**, 6987-6993, 2022.

47. "Large-scale synthesis of mixed valance K₃[Fe₂S₄] with high dielectric and ferrimagnetic characteristics": M. R. Ghazanfari, A. Santhosh, J. C. Vrijmoed, K. Siemensmeyer, B. Peters, S. Dehnen, P. Jerabek, G. Thiele, *RSC Adv.*, **12**, 30514–30521, 2022.
46. "Large Exchange Bias, High Dielectric Constant, and Outstanding Ionic Conductivity in a Single-Phase Spin Glass": M. R. Ghazanfari, A. Santhosh, K. Siemensmeyer, F. Fuß, L. Staab, J. C. Vrijmoed, B. Peters, M. Liesegang, S. Dehnen, O. Oeckler, P. Jerabek, G. Thiele, *Adv. Electron. Mater.*, **8**, 2200483, 2022.
45. "From the gas phase to the solid state: The chemical bonding in the superheavy element flerovium": E. Florez, O. R. Smits, J.-M. Mewes, P. Jerabek, P. Schwerdtfeger, *J. Chem. Phys.*, **157**, 64304, 2022.
44. "An Effective Activation Method for Industrially Produced TiFeMn Powder for Hydrogen Storage": D. M. Dreistadt, L. Thi-Thu, G. Capurso, J. M. Bellosta von Colbe, A. Santhosh, C. Pistidda, N. Scharnagl, H. Ovri, C. Milanese, P. Jerabek, T. Klassen, J. Jepsen, *J. Alloys Compd.*, **919**, 165847, 2022.
43. "Quantitative assessment of ligand substituent effects on σ - and π -contributions to Fe-N bonds in spin crossover FeII complexes": L. Bondi, A. L. Garden, F. Totti, P. Jerabek, S. Brooker, *Chem. Eur. J.*, **28**, e202104314, 2022.
42. "Modeling the thermodynamics of the FeTi hydrogenation under para-equilibrium: An ab-initio and experimental study": E. Alvares, P. Jerabek, Y. Shang, A. Santhosh, C. Pistidda, T. W. Heo, B. Sundman, M. Dornheim, *Calphad*, **77**, 102426, 2022.
41. "Hydrogenation via a low energy mechafalsechemical approach: The MgB₂ case": C. Pistidda, A. Santhosh, P. Jerabek, Y. Shang, A. Girella, C. Milanese, M. Dore, S. Garroni, S. Bordigfalsen, M. R. Chierotti, T. Klassen, M. Dornheim, *J. Phys. Energy*, **3**, 44001, 2021.
40. "Oganesson: A Noble Gas Element That Is Neither Noble Nor a Gas": O. R. Smits, J. Mewes, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **59**, 23636–23640, 2020.
39. "First-principles melting of krypton and xenon based on many-body relativistic coupled-cluster interaction potentials": O. R. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Phys. Rev. B*, **101**, 104103, 2020.
38. "Catalytic Asymmetric Fluorination of Copper Carbene Complexes: Preparative Advances and a Mechanistic Rationale": M. Buchsteiner, L. Martinez-Rodriguez, P. Jerabek, I. Pozo, M. Patzer, N. Nöthling, C. W. Lehmann, A. Fürstner, *Chem. Eur. J.*, **26**, 2509–2515, 2020.
37. "Quantitative and Chemically Intuitive Evaluation of the Nature of M–L Bonds in Paramagnetic Compounds: Application of EDA-NOCV Theory to Spin Crossover Complexes": L. Bondi, A. L. Garden, P. Jerabek, F. Totti, S. Brooker, *Chem. Eur. J.*, **26**, 13677–13685, 2020.
36. "Ligands Based on Phosphine-Stabilized Aluminum(I), Boron(I), and Carbon(0)": L. Vondung, P. Jerabek, R. Langer, *Chem. Eur. J.*, **25**, 3068–3076, 2019.
35. "Oganesson Is a Semiconductor: On the Relativistic Band-Gap Narrowing in the Heaviest falseble-Gas Solids": J.-M. Mewes, P. Jerabek, O. R. Smits, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **58**, 14260–14264, 2019.
34. "Solid Oganesson via a Many-Body Interaction Expansion Based on Relativistic Coupled-Cluster Theory and from Plane-Wave Relativistic Density Functional Theory": P. Jerabek, O. R. Smits, J.-M. Mewes, K. A. Peterson, P. Schwerdtfeger, *J. Phys. Chem. A*, **123**, 4201–4211, 2019.
33. "Dative and Electron-Sharing Bonding in Transition Metal Compounds": P. Jerabek, P. Schwerdtfeger, G. Frenking, *J. Comput. Chem.*, **40**, 247–264, 2019.

32. "A Hundred-Year-Old Experiment Re-evaluated: Accurate Ab-Initio Monte-Carlo Simulations of the Melting of Radon": O. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **57**, 9961–9964, 2018.
31. "The Light-Driven Isomerization of Aqueous Nitrate: A Theoretical Perspective": J.-M. Mewes, P. Jerabek, D. S. Bohle, P. Schwerdtfeger, *ChemPhotoChem*, **2**, 725–733, 2018.
30. "Tipping the Balance between Ligand and Metal Protonation due to Relativistic Effects: Unusually High Proton Affinity in Gold(I) Pincer Complexes": P. Jerabek, L. Vondung, P. Schwerdtfeger, *Chem. Eur. J.*, **24**, 6047–6051, 2018.
29. "A relativistic coupled-cluster interaction potential and rovibrational constants for the xenon dimer": P. Jerabek, O. Smits, E. Pahl, P. Schwerdtfeger, *Mol. Phys.*, **116**, 1–8, 2018.
28. "Static dipole polarizability of palladium from relativistic coupled-cluster theory": P. Jerabek, P. Schwerdtfeger, J. K. Nagle, *Phys. Rev. A*, **98**, 12508, 2018.
27. "Electron and nucleon localization functions of oganesson: Approaching the Thomas-Fermi limit": P. Jerabek, B. Schuettrumpf, P. Schwerdtfeger, W. Nazarewicz, *Phys. Rev. Lett.*, **120**, 53001, 2018.
26. "Suppressed Phosphine Dissociation by Polarization Effects on the Donor–Acceptor Bonds in $[\text{Ni}(\text{PEt}_3)_{4-n}(\text{ECp}^*)_n]$ ($\text{E} = \text{Al}, \text{Ga}$)": J. Hornung, J. Weßing, P. Jerabek, C. Gemel, A. Pöthig, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **57**, 12657–12664, 2018.
25. "Influence of Relativistic Effects on Bonding Modes in M(II) Dinuclear Complexes ($\text{M} = \text{Au}, \text{Ag}, \text{and Cu}$)": P. Jerabek, B. von der Esch, H. Schmidbaur, P. Schwerdtfeger, *Inorg. Chem.*, **56**, 14624–14631, 2017.
24. "The Oxygen-Rich Beryllium Oxides BeO_4 and BeO_6 ": Q. Zhang, P. Jerabek, M. Chen, M. Zhou, G. Frenking, *Angew. Chem. Int. Ed.*, **55**, 10863–10867, 2016.
23. "Understanding the F 1s NEXAFS Dichroism in Fluorinated Organic Semiconductors": M. Klues, P. Jerabek, T. Breuer, M. Oehzelt, K. Hermann, R. Berger, G. Witte, *J. Phys. Chem. C*, **120**, 12693–12705, 2016.
22. "Zn...Zn interactions at nickel and palladium centers": K. Freitag, M. Molon, P. Jerabek, K. Dilchert, C. Rösler, R. W. Seidel, C. Gemel, G. Frenking, R. A. Fischer, *Chem. Sci.*, **7**, 6413–6421, 2016.
21. "The σ -Aromatic Clusters $[\text{Zn}_3]^+$ and $[\text{Zn}_2\text{Cu}]$: Embryonic Brass": K. Freitag, C. Gemel, P. Jerabek, M. I. Opperl, R. W. Seidel, G. Frenking, H. Bann, K. Dilchert, R. A. Fischer, *Angew. Chem. Int. Ed.*, **54**, 4370–4374, 2015.
20. "Stepwise Synthesis of Siloxane-Substituted Oligoarsanes and Structural Investigation of Alkaline Earth Metal Derivatives": C. Clobes, P. Jerabek, I. Nußbruch, G. Frenking, C. von Hänisch, *Eur. J. Inorg. Chem.*, 3264–3273, 2015.
19. "Isolation of Neutral Mofalsenuclear Copper Complexes Stabilized by Two Cyclic (Alkyl)(amino)carbenes": D. S. Weinberger, N. Amin SK, K. C. Mondal, M. Melaimi, G. Bertrand, A. C. Stückl, H. W. Roesky, B. Dittrich, S. Demeshko, B. Schwederski, W. Kaim, P. Jerabek, G. Frenking, *J. Am. Chem. Soc.*, **136**, 6235–6238, 2014.
18. "A Novel Concept for the Synthesis of Multiply Doped Gold Clusters $[(\text{M}@\text{Au}_n\text{M}'_m)\text{Lk}]_q^+$ ": A. Puls, P. Jerabek, W. Kurashige, M. Molon, T. Bollermann, M. Winter, C. Gemel, Y. Negishi, G. Frenking, R. A. Fischer, *Angew. Chem. Int. Ed.*, **53**, 4327–4331, 2014.
17. "Experimental Charge Density Study of a Silylone": B. Niepötter, R. Herbst-Irmer, D. Kratzert, P. P. Samuel, K. C. Mondal, H. W. Roesky, P. Jerabek, G. Frenking, D. Stalke, *Angew. Chem. Int. Ed.*, **53**, 2766–2770, 2014.

16. "Hume-Rothery Phase-Inspired Metal-Rich Molecules: Cluster Expansion of $[\text{Ni}(\text{ZnMe})_6(\text{ZnCp}^*)_2]$ by Face Capping with $\text{Ni}(0)(\eta(6)\text{-toluene})$ and $\text{Ni}(\text{I})(\eta(5)\text{-Cp}^*)$ ": M. Molon, C. Gemel, P. Jerabek, L. Trombach, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **53**, 10403–10411, 2014.
15. "Coinage Metals Binding as Main Group Elements: Structure and Bonding of the Carbene Complexes $[\text{TM}(\text{cAAC})_2]$ and $[\text{TM}(\text{cAAC})_2]^+$ (TM = Cu, Ag, Au)": P. Jerabek, H. W. Roesky, G. Bertrand, G. Frenking, *J. Am. Chem. Soc.*, **136**, 17123–17135, 2014.
14. "Comparative bonding analysis of N_2 and P_2 versus tetrahedral N_4 and P_4 ": P. Jerabek, G. Frenking, *Theor. Chem. Acc.*, **133**, 1447–1456, 2014.
13. "Bonding analysis of telluroketones $\text{H}_2\text{A}=\text{Te}$ (A = C, Si, Ge)": N. B. Jaufeerally, P. Ramasami, P. Jerabek, G. Frenking, *J. Mol. Mod.*, **20**, 2433–2440, 2014.
12. "Dizinc Cation $[\text{Zn}_2]^2+$ Trapped In A Homoleptic Metalloid Coordination Environment Stabilized by Dispersion Forces: $[\text{Zn}_2(\text{GaCp}^*)_6][\text{BAr}_4\text{F}]^2$ ": K. Freitag, H. Banh, C. Gemel, P. Jerabek, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **54**, 352–358, 2014.
11. "Isolation of Neutral Mofalse- and Dinuclear Gold Complexes of Cyclic (Alkyl)(amino)carbenes": D. S. Weinberger, M. Melaimi, C. E. Moore, A. L. Rheingold, G. Frenking, P. Jerabek, G. Bertrand, *Angew. Chem. Int. Ed.*, **52**, 8964–8967, 2013.
10. "The Orgafalsezinc Rich Compounds $[\text{Cp}^*\text{M}(\text{ZnR})_5]$ (M = Fe, Ru; R = Cp^* , Me, Cl, Br)": M. Molon, C. Gemel, R. W. Seidel, P. Jerabek, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **52**, 7152–7160, 2013.
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